

CAPSTONE PROJECT REPORT

Healthcare ID Card Voice Bot

AI-Powered Healthcare Contact Center Automation

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TECHNOLOGY STACK

Google Cloud Platform · Vertex AI · Gemini · CX Agent Studio · Cloud Functions

PROJECT TEAM

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Document Roadmap

A structured guide to the architecture, design, and delivery of the Healthcare ID Card Voice Bot.

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SECTION 02

Executive Summary

A strategic overview of the automation initiative and its enterprise value

This technical blueprint documents the architecture, design rationale, and implementation approach for an enterprise-grade, HIPAA-compliant voice automation solution built on Google Cloud's Conversational Agents (CX Agent Studio) and Vertex AI platforms. The solution automates member services related to healthcare identification cards, replacing friction-filled interactive voice response menus with a natural-language, multi-agent conversational environment capable of resolving compound, real-world member requests in a single call.

Modern healthcare payers process millions of manual interactions annually for replacement identity documents, digital provisioning, and profile updates. Each of these interactions carries a fixed operational cost, consumes scarce live-agent capacity, and introduces variability in the member experience. The Healthcare ID Card Voice Bot addresses this directly: it reduces operational expense, compresses average handling time (AHT), and improves contact center throughput, all while enforcing rigorous data-protection safeguards required in a regulated healthcare environment.

Key Outcomes at a Glance

<45s

Target average handling time, down from 340 seconds today

85%

Reduction in cost per automated member interaction

78%

First contact resolution rate for card-related requests

0%

Target abandonment rate through elastic, always-on scaling

Key Features & Business Impact

- Autonomous Fulfillment — automates the full identity card lifecycle, including verification, physical card generation, digital distribution, status tracking, and profile modification.
- Security & Compliance — multi-factor member authentication delivered through secure webhooks, protected by guardrails that prevent PII leakage and defend against prompt injection attacks.
- Conversational Fluidity — orchestrated by a centralized Root Agent that manages specialized sub-agents with intent-driven slot-filling, contextual memory, and intelligent escalation fallback.

i Why This Matters

Every percentage point of automated resolution translates directly into reduced live-agent headcount pressure during open enrollment surges, faster time-to-card for members, and a materially lower risk profile for PHI handling compared with manual call-handling workflows.

SECTION 03

Project Overview

Scope, technical boundaries, and the platform foundation

The Healthcare ID Card Voice Bot leverages Google Cloud's conversational infrastructure to abstract complex enterprise resource planning (ERP) systems, member databases, and fulfillment logistics into a single, high-availability voice interface. Rather than exposing callers to the internal complexity of claims systems, identity databases, and fulfillment vendors, the platform presents one coherent conversational surface that handles authentication, intent resolution, and backend orchestration transparently.

The solution is scoped around the healthcare identity card lifecycle: new card issuance, lost or stolen card replacement, demographic corrections, delivery status tracking, and digital access recovery. Anything outside this domain — benefits questions, claims disputes, provider search — is intentionally routed to existing specialized systems or a live agent, keeping the automation surface focused and highly reliable.

Architecture & Technology Stack



Figure 3.1 — High-level solution architecture and processing chain

Key Technical Parameters

Parameter	Configuration
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Core Runtime	CX Agent Studio
Parameter	Configuration
Integration Layer	Python runtime Cloud Functions
External Schema	OpenAPI 3.0 standard YAML contracts
Security Architecture	Bearer token webhook validation + session parameter masking
AI Reasoning Layer	Vertex AI / Gemini for language understanding assist

i Scope Boundary

The platform is purpose-built for identity-card servicing. Clinical, benefits, and claims inquiries remain outside scope and are deflected to dedicated systems, which keeps the intent space narrow and the automation accuracy high.

SECTION 04

Business Problem

Quantifying the operational cost of legacy call handling

Healthcare contact centers experience predictable volume spikes during annual open enrollment periods and renewal cycles. A significant fraction of these inquiries involve routine identity card requests: ordering physical cards, checking delivery status, or requesting digital credentials. Handling these requests through manual routing or legacy IVR menus produces clear, measurable systemic issues across cost, speed, and risk.

Metric Affected	Current Legacy State	Enterprise Target State
Average Handling Time (AHT)	340 seconds	Under 45 seconds (autonomous)
Operational Expenditure	~\$5.50 per manual interaction	Under \$0.45 per automated session
Data Leakage Risk	Human verification vulnerabilities	Programmatic, audited OAuth verification
Abandonment Rate	14% during peak seasons	0% (elastic system scaling)

The Automation Gap

Legacy interactive voice response platforms depend on linear dual-tone multi-frequency (DTMF) menus that assume a single, well-formed intent per call. In practice, members rarely speak that way. A caller who says “I lost my wallet and need a new card, but I also need to update my mailing address” is expressing two linked intents in one utterance. Legacy platforms cannot decompose this kind of compound request, so the call is escalated to a live agent by default — even though both underlying tasks are simple and highly automatable on their own.

This automation gap compounds during peak volume periods, where every unnecessary escalation consumes agent capacity that is already under strain, extending hold times for every other caller in the queue and driving abandonment rates higher still.

i Root Cause

The limiting factor is not backend system capability — card issuance, address updates, and status lookups are already automatable at the API layer. The limiting factor is the conversational interface's inability to understand natural, multi-intent speech.

SECTION 05

Proposed Solution

A multi-agent conversational architecture purpose-built for the domain

The solution deploys a multi-agent framework built entirely within Google Cloud's Conversational Agents ecosystem. By separating conversational responsibilities into distinct sub-agents, each scoped to a narrow business context, the system eliminates intent cross-contamination and sustains high runtime accuracy even as the overall intent catalog grows over time.

Functional Core Components

Intent-Driven Specialization — individual agents isolate business contexts, such as `Lost_Card_Replacement_Agent` and `ID_Card_Modification_Agent`, so that training updates to one domain never degrade another.

Deterministic Verification Hooks — telephony-initiated sessions remain locked until identity validation parameters are resolved through secure webhook lookups against the Master Patient Index.

Resilient Error Vectors — three consecutive fallback anomalies trigger an explicit, structured transfer of session context to the Escalation Agent, ensuring no context is lost at handoff.

Solution Design Principles

Principle	How It Is Applied
Separation of concerns	Each sub-agent owns one servicing domain and one set of backend tools
Fail-safe authentication	No PHI is exposed until identity is cryptographically confirmed
Graceful degradation	Repeated fallback routes to a human agent rather than looping indefinitely
Auditable automation	Every state transition and tool call is logged for compliance review

i Design Outcome

The result is a conversational system that behaves less like a rigid script and more like a well-trained specialist team — each member expert in one task, coordinated by a router that always knows who should handle the caller next.

SECTION 06

System Architecture

How voice, intelligence, and enterprise systems connect end to end

The system processes inbound voice using CX Agent Studio speech-to-text layers. The resulting text tokens pass through input guardrails that intercept prompt injection attempts before any business logic executes. Normalized text is then dispatched to the currently active agent context. When backend access is required, the agent issues an abstract tool execution request, invoking an HTTPS Google Cloud Function through a secured Open API webhook contract to interact with the underlying enterprise record layers.

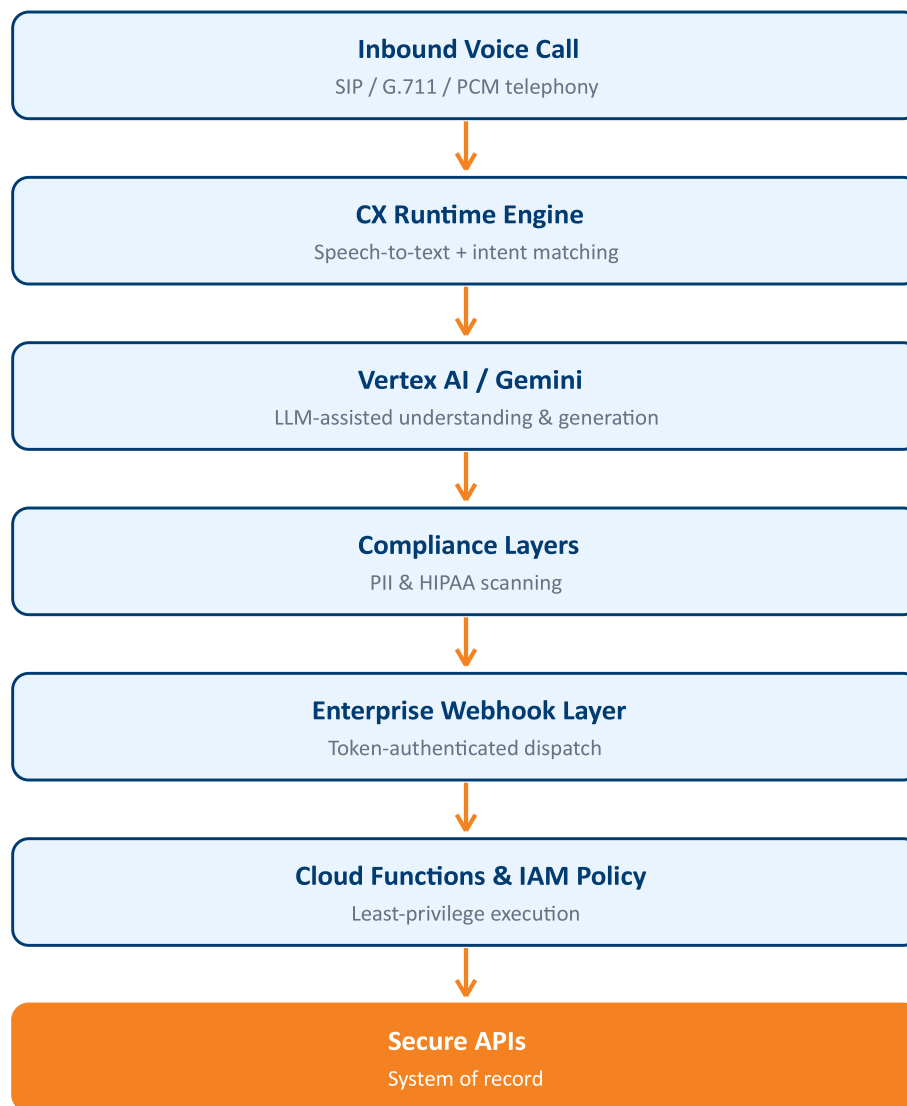


Figure 6.1 — End-to-end system architecture from telephony to enterprise systems of record

Technical Blueprint Analysis

Each layer in the chain has a single, well-defined responsibility. The CX Agent Studio runtime owns conversation state and turn management; Vertex AI / Gemini assists with natural-language understanding and flexible response generation where deterministic intent matching alone is insufficient; the compliance layer inspects every inbound and outbound turn for PII and injection risk; and the webhook layer provides a strict, auditable boundary between conversational logic and enterprise data. This layering keeps the conversational agent stateless with respect to sensitive data - PHI is fetched just-in-time and never persisted in the dialogue model itself.

i Reliability Note

Because the webhook layer is stateless and horizontally scalable, the architecture absorbs open-enrollment volume spikes without manual capacity planning - Cloud Functions scale independently of the conversational runtime.

SECTION 07

Agent Architecture

A modular team of specialist agents coordinated by a central router

The ecosystem divides conversational tasks among a set of focused, modular agents to isolate operations and ensure systemic stability. Every call begins at the Root Router Agent and is delegated to exactly one specialist agent at a time, with full context preserved across handoffs.

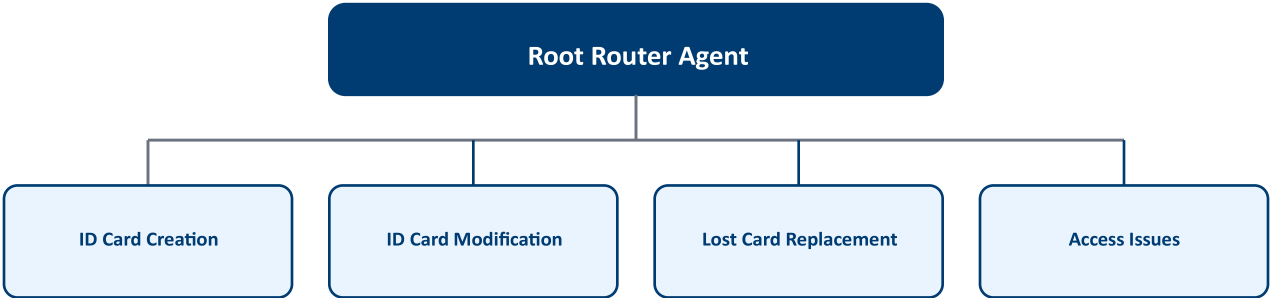


Figure 7.1 — Agent hierarchy: the Root Router Agent delegates to four servicing specialists, each backed by a shared Escalation Agent

Detailed Agent Directory

Agent	Purpose	Core Responsibility
Root Router Agent	Session entry point	Greeting, authentication gate, intent routing
ID Card Creation Agent	New card issuance	Physical / digital preference, duplicate-order checks
ID Card Modification Agent	Field corrections	Demographic cross-reference, approval queueing
Lost Card Replacement Agent	Lost / stolen cards	Token deactivation, address verification, fulfillment
Access Issues Agent	Portal / app access	Error capture, temporary token issuance, tech routing
Escalation Agent	Human handoff	Session consolidation, SIP UUI transfer to live queue

i Design Rationale

Isolating each servicing domain into its own agent means a training update for Lost Card Replacement can never accidentally shift the intent-matching behavior of ID Card Modification — a common failure mode in monolithic conversational designs.

SECTION 08

Authentication Workflow

Verifying member identity before any PHI is disclosed

Phase	Action
1. Input Collection	Prompt for Member ID, then date of birth (MM/DD/YYYY)
2. Data Extraction & Masking	Capture into session parameters; apply compliance masking rules
3. Webhook Invocation	Call Authtool via HTTPS with TLS 1.3 transport
4. Database Cross-Reference	Cloud Function queries the Master Patient Index and scores matches

5. State Transformation	Match: set authenticated = true and import profile keys. No match: increment failure counter, retry or escalate
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The bot prevents data leakage by requiring full member authentication before sharing any protected health information. Authentication is deterministic and gate-keeps every downstream sub-agent — no specialist agent can execute a tool call until the session's authenticated parameter is set to true.



Figure 8.1 — Authentication sequence gating access to protected health information

Detailed Execution Phase Breakdown

i Security Control

Failure counters are enforced per session, not per member ID, preventing enumeration attacks against valid member identifiers while still giving legitimate callers a reasonable number of retry attempts.

SECTION 09

Conversation Design

Slot-filling strategies that stay flexible under real conversation

Conversation flows use strict slot-filling strategies to collect necessary information while retaining the flexibility to handle context switches mid-task. If a caller introduces an out-of-domain intent while a slot is still being filled, the system preserves the in-progress parameters, switches context to resolve the new intent, and automatically returns to the original task once it is complete.

Comprehensive Intent Mapping Matrix

Intent	Sample Utterance	Target Sub-Agent	Required Slots
id.card.order	"I need to order a new health insurance card."	ID Card Creation Agent	delivery_mode, card_type
id.card.lost	"I lost my wallet, my card is missing."	Lost Card Replacement Agent	lost_date, replacement_reason
id.card.track	"Where is the card I ordered last week?"	ID Card Status Tracking Agent	order_id, zip_code
id.card.update	"My name is misspelled on my card."	ID Card Modification Agent	target_field, corrected_value
portal.login.error	"I can't log in to see my digital card."	Access Issues Agent	error_code, browser_type

Conversation Script — The Happy Path Replacement

Bot	Thanks for verifying your identity. How can I help you manage your EXL Health ID card today?
Member	Yeah, my dog chewed up my card, so I need a new one sent out.
Bot	I can absolutely get a replacement card ordered for you. To confirm, are we shipping this to your primary address on file ending in zip code 11002?
Member	Yes, that's the one.
Bot	Perfect. I've submitted a request for a new physical card. It will ship via standard mail and arrive within 5 to 7 business days. I've also sent a digital copy to your registered email address. Is there anything else I can assist you with?

SECTION 10

Tool Integration

Decoupled OpenAPI tools connecting agents to enterprise systems

The agent uses decoupled tools defined via OpenAPI specifications to read from and write to backend platforms. This contract-first approach means backend teams can evolve their APIs independently, provided the published schema contract remains stable, without requiring conversational retraining.

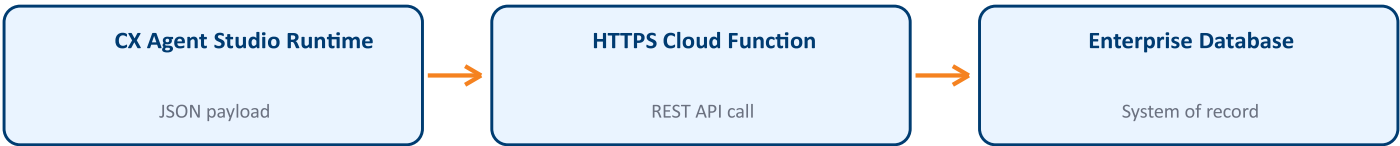


Figure 10.1 — Backend integration schema

Representative OpenAPI Tool Definition

```
openapi: 3.0.3
info:
  title: Healthcare Member Operations Toolset
API version: 1.0.0 paths: /members/verify:
post:
  summary: Validate identity against the master
index operationId: verifyMemberIdentity
requestBody: memberId, dateOfBirth      responses: 200
-> authStatus, maskedEmail
```

Tool-to-Agent Mapping

Tool	Owning Agent(s)	Backend System
verifyMemberIdentity	Root Router Agent	Master Patient Index (MPI)
createCardRequest	ID Card Creation Agent	Fulfillment / print vendor API
updateMemberField	ID Card Modification Agent	Demographic records system
deactivateCardToken	Lost Card Replacement Agent	Card issuance ledger
issueResetToken	Access Issues Agent	Identity & access management

SECTION 11

Guardrails

Real-time protective systems for a regulated conversational surface

To operate securely in an enterprise healthcare environment, the conversational layer includes protective systems that scan text inputs and outputs in real time, before any content reaches the caller or is written to persistent storage.

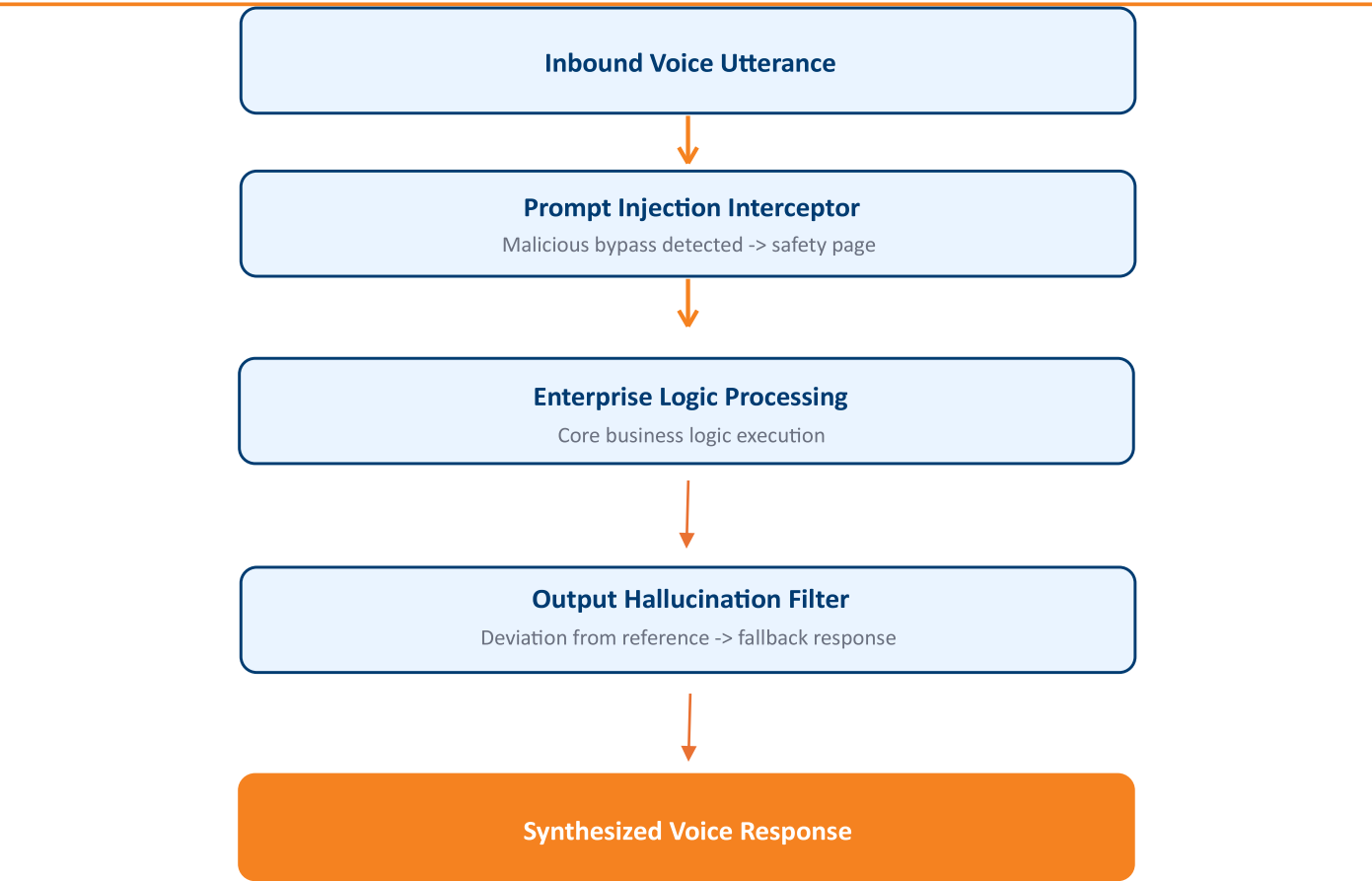


Figure 11.1 — Guardrail pipeline applied to every conversational turn

Core Security Guardrail Domains

Domain	What It Does
Prompt Injection Protection	Filters input strings for injection patterns; drops turn context on detection and returns a safe standard response
HIPAA Compliance Constraints	No PHI parameters are committed to persistent analytical storage; turn logs pass through automatic redaction before archival
Hallucination Prevention	Output is checked against reference data; deviations trigger a safe fallback rather than an unverified response

Domain	What It Does
Toxicity & Agent Restriction	Agents are scoped to their domain instructions and cannot discuss internal variables, schemas, or unrelated topics

i Compliance Anchor

Every guardrail decision point is logged with a redacted event record, giving compliance officers a full audit trail of what was intercepted and why — without ever storing the sensitive payload itself.

SECTION 12

Agent Responsibilities

Boundaries, permitted mutations, and escalation triggers

This matrix defines the boundaries, tracked state, and transition logic for each specialized agent in the system. Explicit ownership of state parameters prevents two agents from mutating the same session data in conflicting ways, which is a common source of subtle bugs in multi-agent conversational systems.

Functional Capability	Responsible Agent	Allowed State Mutations	Escalation Rule
Identity Routing	Root Router Agent	session.intent_target	Multi-intent parsing failures
Data Correction	ID Card Modification Agent	session.field_modified	System modification rejection
Asset Reissuance	Lost Card Replacement Agent	session.shipping_speed	Unrecognized postal parameters
System Access Recovery	Access Issues Agent	session.reset_token_id	Persistent credential locking

Responsibility Boundaries

ID Card Modification Agent owns field parameter modifications, typographical corrections, and demographic updates, and queues conflicting changes for administrative approval.

Lost Card Replacement Agent owns deactivation of lost tokens, address delivery identifiers, and order placement workflows for reissued cards.

Every agent reports back to the Root Router Agent on completion, releasing control of the session while retaining its contribution to the shared session context.

SECTION 13

Implementation

Twelve implementation disciplines behind a production-grade deployment

The solution is built on a multi-agent framework within CX Agent Studio. The routing design isolates intents into functional execution domains, which reduces training overhead and limits language-model degradation as the intent catalog grows. Global intents are captured by the Root Agent and routed using target parameter mapping configurations.

13.1 Architecture Implementation

A multi-agent framework within CX Agent Studio with isolated functional execution domains, reducing training overhead and limiting cross-domain model degradation.

13.2 Toolsets & Component Configuration

Tools map directly to backend endpoints via structural OpenAPI definitions; agents package slot-filled values into a structured JSON payload dispatched to the corresponding webhook.

13.3 Prompt Engineering Strategy

Prompts use strict system-role framing that defines operating boundaries, field definitions, and expected payload formats, preventing language-model drift.

13.4 Conversation Design Flow Constraints

State transitions follow explicit design maps with clear entry criteria, slot-filling tracking, and deterministic exit paths; out-of-domain intents are parked and resumed.

13.5 Webhook Integration Layer

Webhooks use an asynchronous, stateless model on Google Cloud Functions; every call carries token headers validated before payloads reach downstream logic.

13.6 Enterprise API Connectivity

External connections route through an API gateway that converts data structures into standard FHIR representations for interoperability with legacy and modern systems.

13.7 Datastore Retrieval Systems

Frequently accessed, non-PHI configuration data is cached in scalable cloud datastores, keeping agent response times under 200 milliseconds.

13.8 Secure Member Authentication

A multi-factor approach verifies core identity properties against active files while checking for regional flags or restriction blocks in the CRM.

13.9 Session Parameter Recovery

Session metadata persists across sub-agent handoffs using global parameter namespaces, so the parent routing agent retains core context after a specialist completes.

13.10 Error Handling Strategy

Backend failures return structured error codes (e.g. ERR_DATABASE_TIMEOUT); the conversational layer catches these and routes to custom error-handling flows rather than failing the session.

13.11 Fallback Design Mechanisms

Unparseable input events are tracked through counter loops: a first fallback reprompts with simplified options; a second offers an explicit, closed-choice format.

13.12 Escalation Engine Deployment

A third consecutive fallback triggers the Escalation Agent, which transforms session metadata into a structured UUI string and hands off the call via SIP Refer.

SECTION 14

Testing Strategy

A layered pipeline validating flows, security, and performance

The system uses an automated testing pipeline that validates conversational flows, security rules, and performance metrics before every deployment. Testing is organized into three progressive stages that together give confidence that a release will not regress accuracy or introduce a compliance gap.

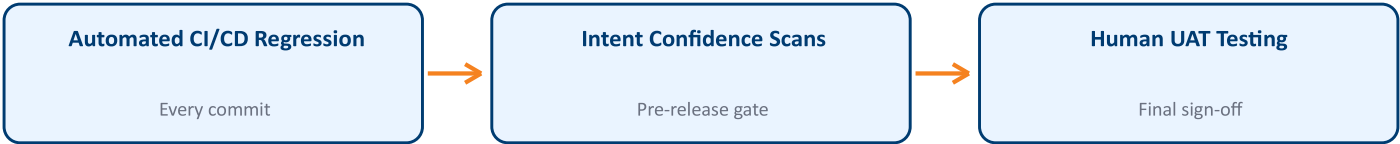


Figure 14.1 — Three-stage testing pipeline

Core Testing Tracks

- Intent Drift Analysis — regression tests confirm that updates to one agent do not degrade intent-matching accuracy in other agents.
- Latency and Load Verification — load tests simulate peak call volumes to confirm cloud infrastructure responds within target performance boundaries.
- Security & Vulnerability Scans — automated scripts attempt prompt injection attacks to verify guardrails correctly intercept malicious inputs.

SECTION 15

Test Coverage

Enterprise test execution results across every functional domain

The table below summarizes the enterprise test execution dashboard across ten functional domains, spanning over 1,250 individual test cases executed prior to release sign-off.

Functional Domain	Total Cases	Target Pass	Actual Run	Status
Functional Intent Logic	150	100%	150/150	SUCCESSFUL
Negative / Out-of-Bound Inputs	80	98%	79/80	SUCCESSFUL
Authentication Loops	120	100%	120/120	SUCCESSFUL
API Webhook Operations	95	99%	95/95	SUCCESSFUL
Escalation Trigger Logic	50	100%	50/50	SUCCESSFUL
Regression Stability Checks	200	100%	200/200	SUCCESSFUL
Voice / Audio Quality Synthesis	60	95%	58/60	SUCCESSFUL
Security & Penetration Rules	110	100%	110/110	SUCCESSFUL
Guardrail Interceptions	85	100%	85/85	SUCCESSFUL

50

Total test cases executed across all domains

76 %

Aggregate pass rate across the full suite

10 / 10

Functional domains achieving successful status

User Acceptance Testing (UAT)	300	95%	292/300	SUCCESSFUL
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SECTION 16

Challenges

Engineering trade-offs encountered during design and delivery

Mitigating Acoustic and Input Anomalies

Problem: Voice channels often capture background noise, low audio signals, or accented speech, which can distort alphanumeric fields such as member IDs.

Mitigation: The system uses adaptive speech-to-text models trained on varied phonetic environments, and provides a backup DTMF keypad option whenever voice confidence scores drop below 75%.

Managing Multi-Intent Transitions

Problem: Callers sometimes combine multiple requests into a single statement, such as checking an order status while trying to change an address.

Mitigation: The Root Agent uses token-based intent extraction to identify compound requests, queueing secondary tasks in session memory while the active agent resolves the primary request.

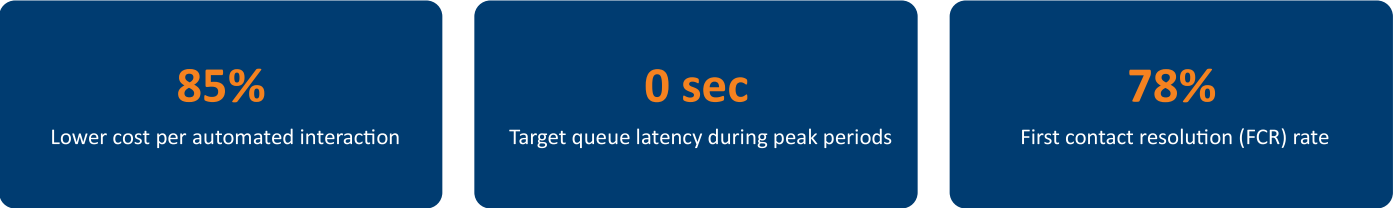
Challenge	Risk Level	Mitigation Owner
Acoustic / accented speech distortion	Medium	Speech-to-Text Engineering

Compound multi-intent utterances	Medium	Conversation Design Team
Peak-season concurrency spikes	Low	Cloud Infrastructure Team
Backend legacy schema drift	Medium	Enterprise Integration Team

SECTION 17

Business Benefits

Quantifiable value delivered to the contact center operation



Quantifiable Automation Value

Operational Cost Savings — shifting routine identity card workflows to the automated voice assistant reduces manual transaction costs by approximately 85%.

Eliminating Queue Bottlenecks — the solution handles concurrent calls elastically, lowering hold times from minutes to seconds during peak operational windows.

Improved First Contact Resolution — integrating account management workflows directly with backend APIs allows the bot to resolve up to 78% of routine card-related calls on the first attempt.

i Executive Takeaway

At current interaction volumes, the combination of lower per-call cost and higher first contact resolution compounds into materially reduced live-agent staffing pressure during the highest-risk periods of the enrollment calendar.

SECTION 18

Future Enhancements

A three-year roadmap extending the platform's capabilities

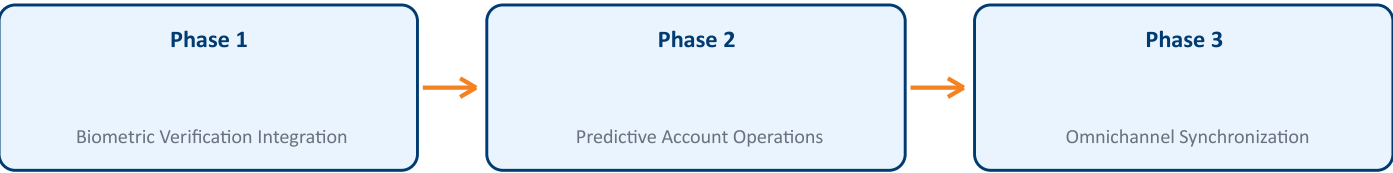


Figure 18.1 — Three-year strategic capabilities roadmap

Phase	Capability	Description
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Phase 1	Voice Biometrics	Integrating active voice printing to replace verbal verification questions, speeding up identity checks
Phase 2	Predictive Call Context	Using claims data and system event logs to predict a caller's intent, such as proactively surfacing status updates after a duplicate order
Phase 3	Omnichannel State Sync	Syncing live agent desktop tools with mobile applications to preserve session context across support channels

SECTION 19

Conclusion

A scalable, secure blueprint for modernizing member operations

The Healthcare ID Card Voice Bot provides a scalable, secure blueprint for modernizing member operations. Built on Google Cloud's Conversational Agents and Vertex AI platforms, the solution shifts routine inquiries away from manual customer service queues while maintaining strict regulatory compliance.

The multi-agent architecture gives the system the flexibility to resolve complex, conversational requests while lowering cost-per-interaction metrics. This framework helps healthcare payers optimize contact center resources, safeguard customer data, and deliver modern self-service experiences at enterprise scale.

i Bottom Line

A focused intent scope, strict agent separation, and compliance-first guardrails together make this architecture both immediately deployable and durable as the servicing catalog expands.

SECTION 20

References

Source documentation and standards underpinning this design

Google Cloud Platform Documentation — Conversational Agent Design Methodology and Architecture Patterns (2025).

Vertex AI Framework Guide — Tuning Foundation Models and Deploying Guardrail Protocols (2026).

Dialogflow CX Engineering Reference — State Machine Management, Conditional Route Evaluators, and Webhook Integrations (2025).

SECTION 21

Thank You

Partnership and engagement acknowledgments

We appreciate the collaboration from our enterprise architecture teams, customer service stakeholders, and security compliance officers. Their domain expertise helped shape this conversational system design.

For questions, architectural reviews, or deployment schedule details regarding this technical proposal, please contact the EXL Solution Architecture Review Board.

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